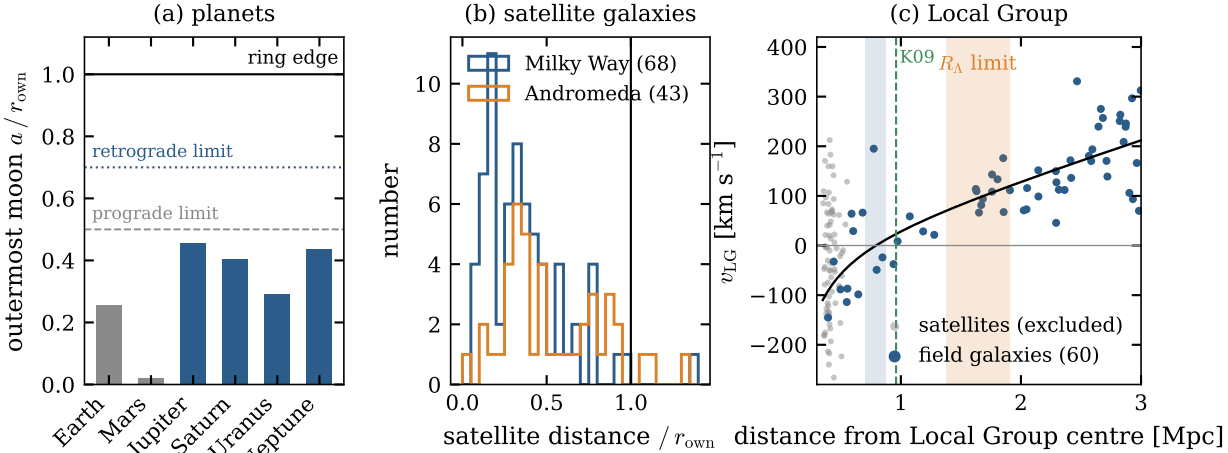
(a) planets
ring edge
retrograde limit
prograde limit
outermost moon $a$ / $r_{\rm own}$
1.0
0.8
0.6
0.4
0.2
0.0
Earth
Mars
Jupiter
Saturn
Uranus
Neptune
(b) satellite galaxies
Milky Way (68)
Andromeda (43)
number
0
2
4
6
8
10
0.0
0.5
1.0
satellite distance / $r_{\rm own}$
(c) Local Group
K09
$R_\Lambda$ limit
satellites (excluded)
field galaxies (60)
$v_{\rm LG}$ [km s$^{-1}$]
400
300
200
100
0
−100
−200
1
2
3
distance from Local Group centre [Mpc]